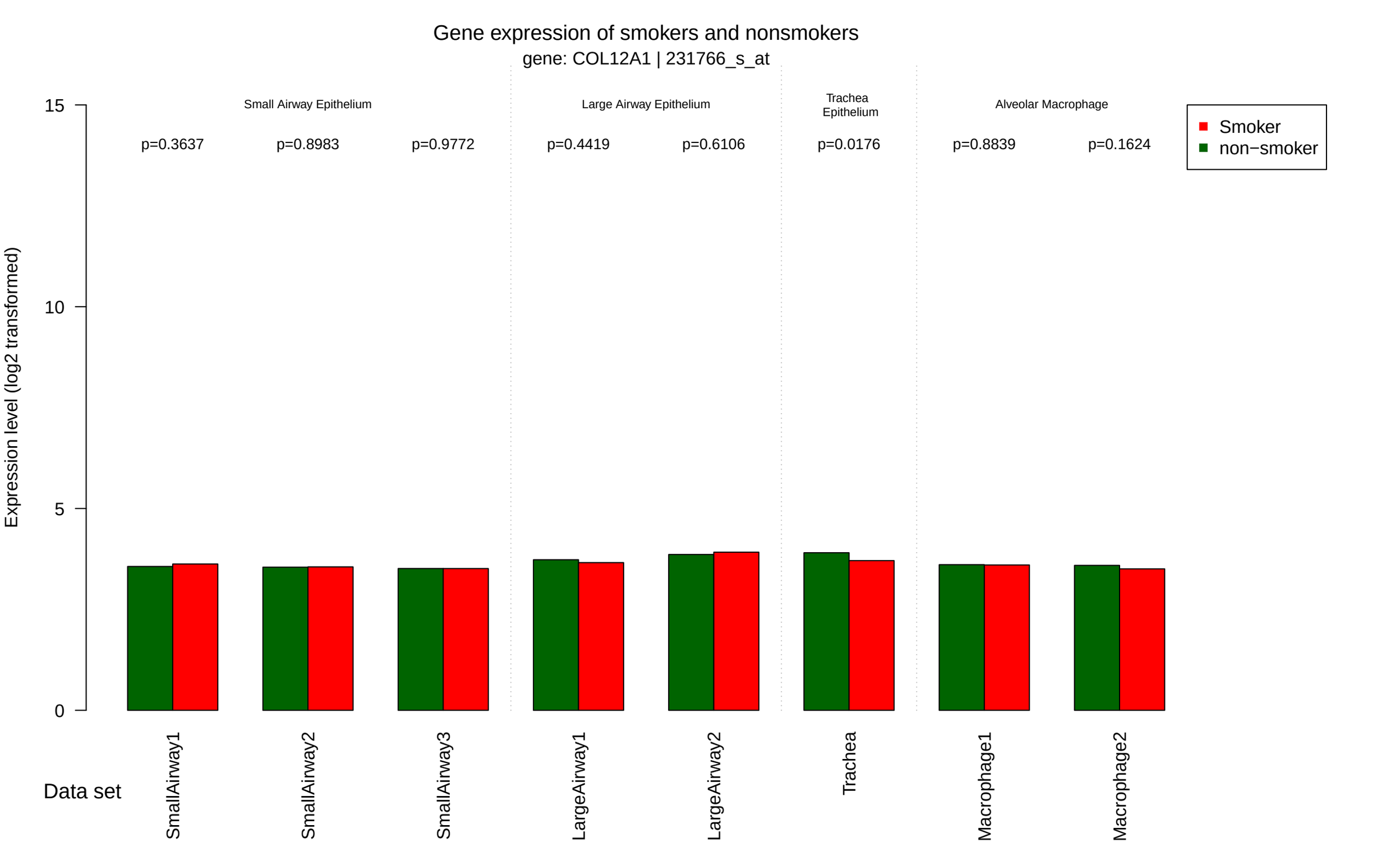
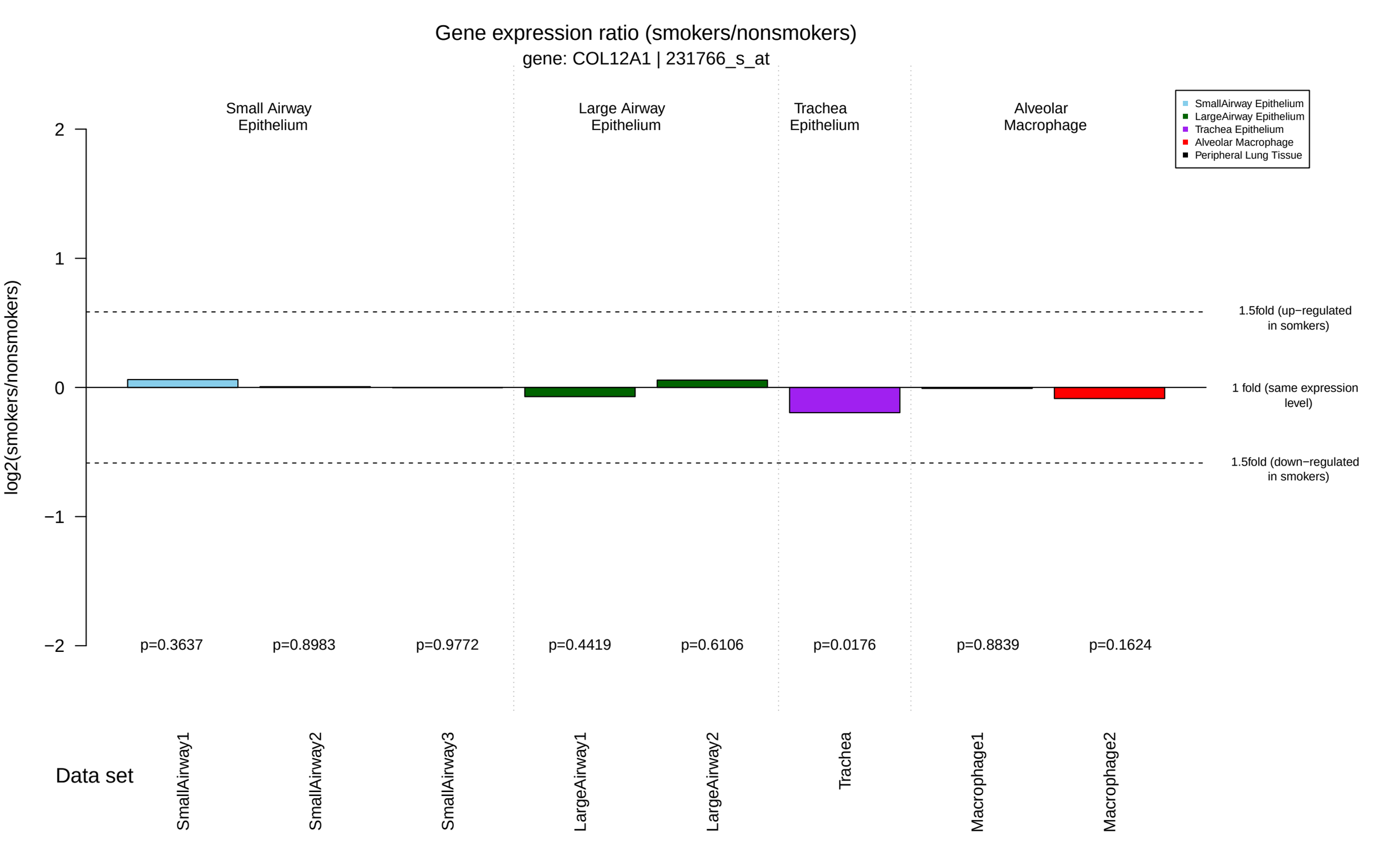
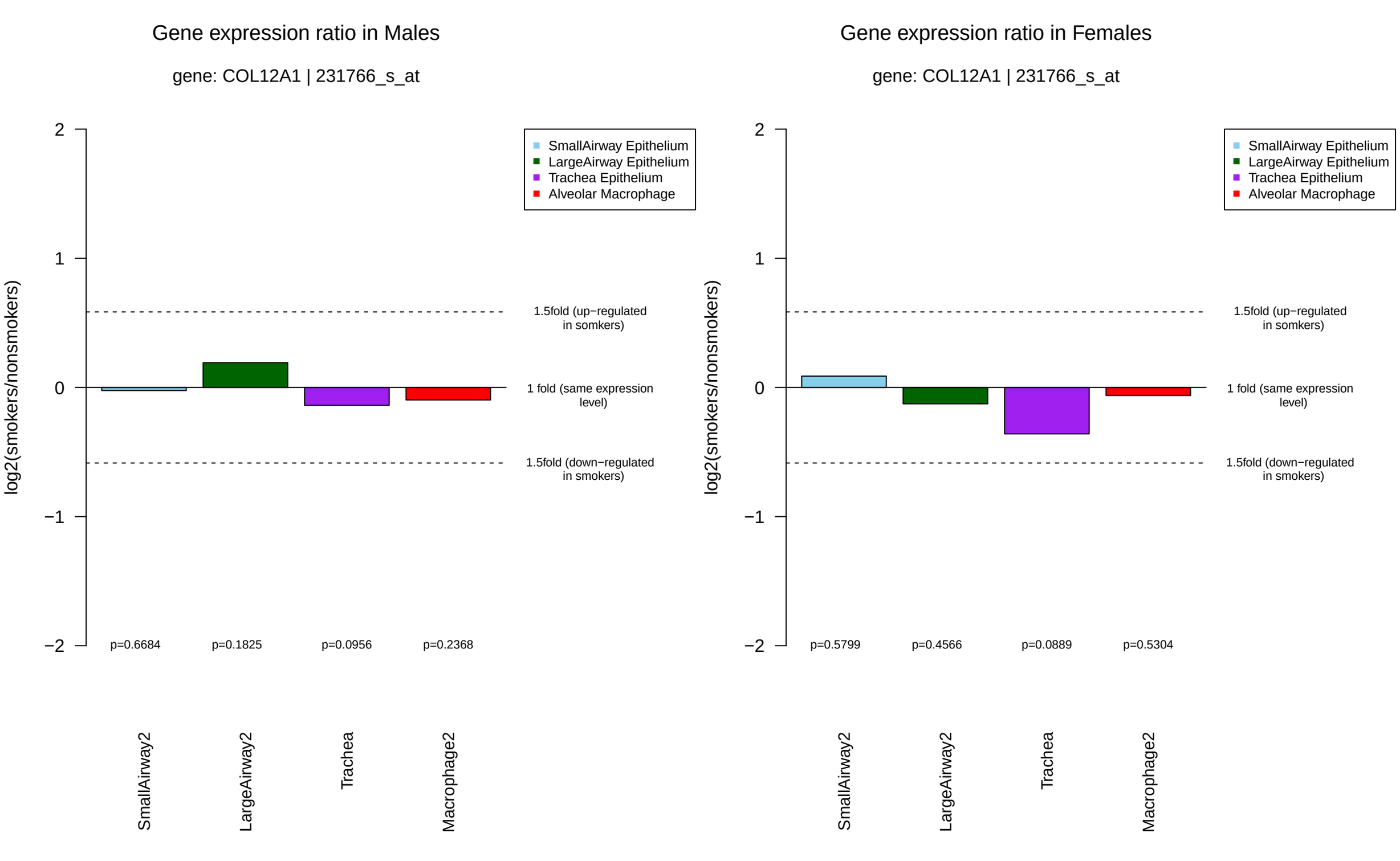
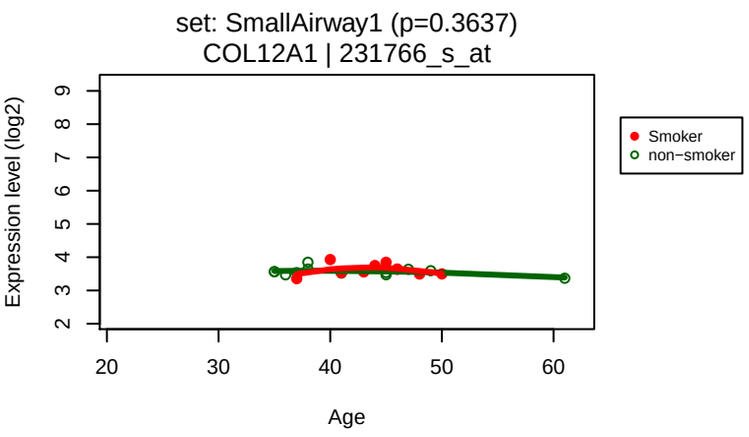
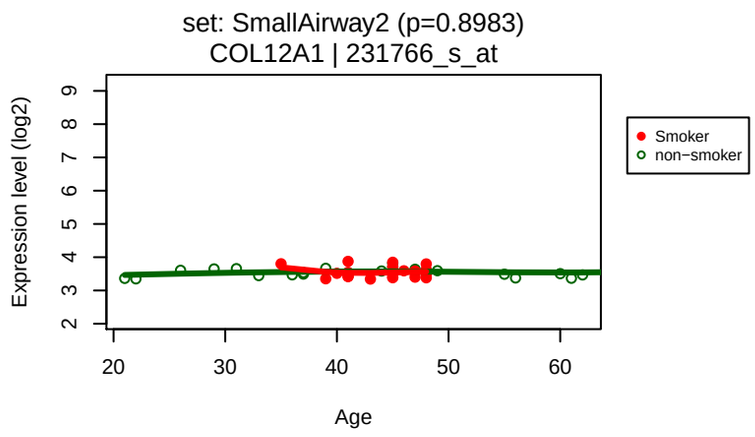
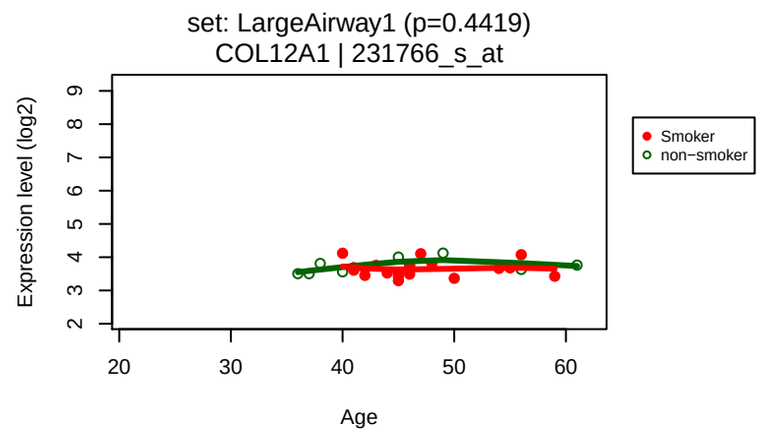

Gene expression of smokers and nonsmokers
gene: COL12A1 | 231766_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3637
p=0.8983
p=0.9772
p=0.4419
p=0.6106
p=0.0176
p=0.8839
p=0.1624
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

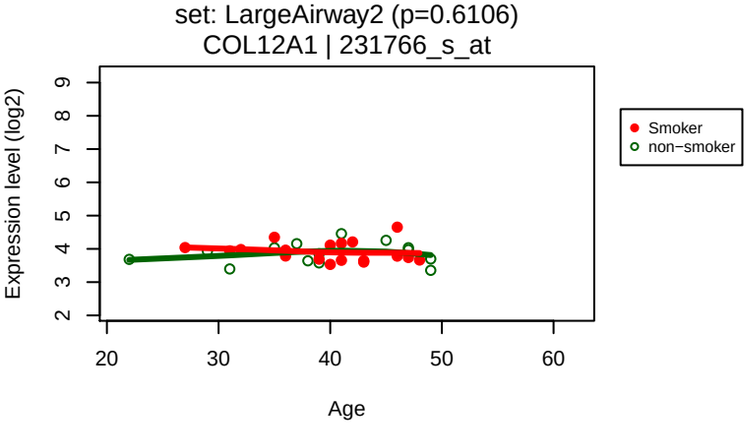

Gene expression ratio (smokers/nonsmokers)
gene: COL12A1 | 231766_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3637
p=0.8983
p=0.9772
p=0.4419
p=0.6106
p=0.0176
p=0.8839
p=0.1624
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

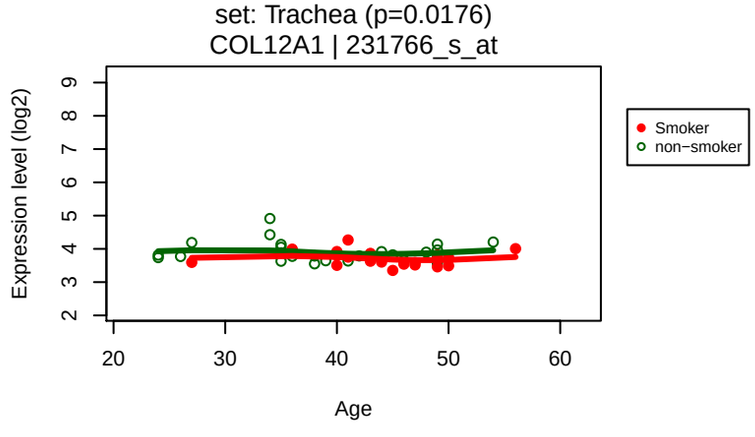

Gene expression ratio in Males
gene: COL12A1 | 231766_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6684
p=0.1825
p=0.0956
p=0.2368
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL12A1 | 231766_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5799
p=0.4566
p=0.0889
p=0.5304
SmallAirway2
LargeAirway2
Trachea
Macrophage2

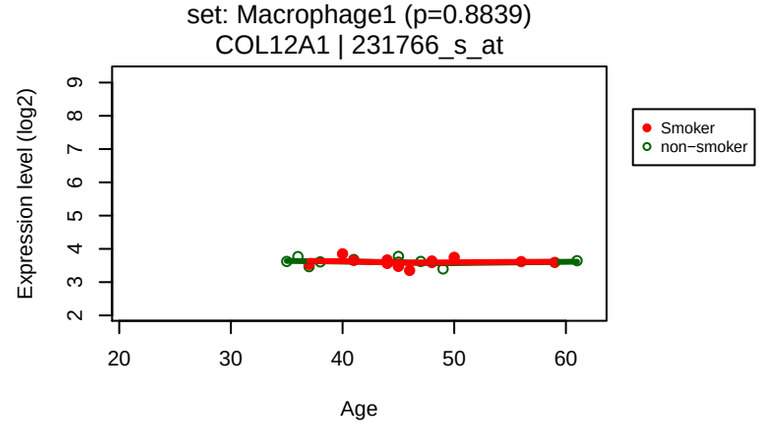

set: SmallAirway1 (p=0.3637)
COL12A1 | 231766_s_at
Expression level (log2)
Age
Smoker
non−smoker

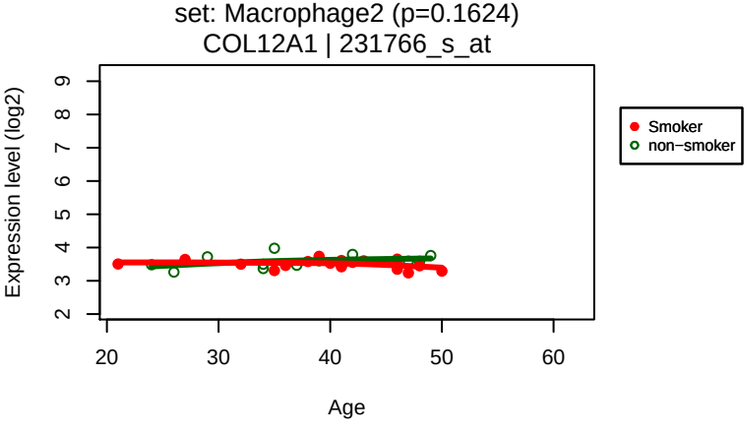

set: SmallAirway2 (p=0.8983)
COL12A1 | 231766_s_at
Expression level (log2)
Age
Smoker
non−smoker

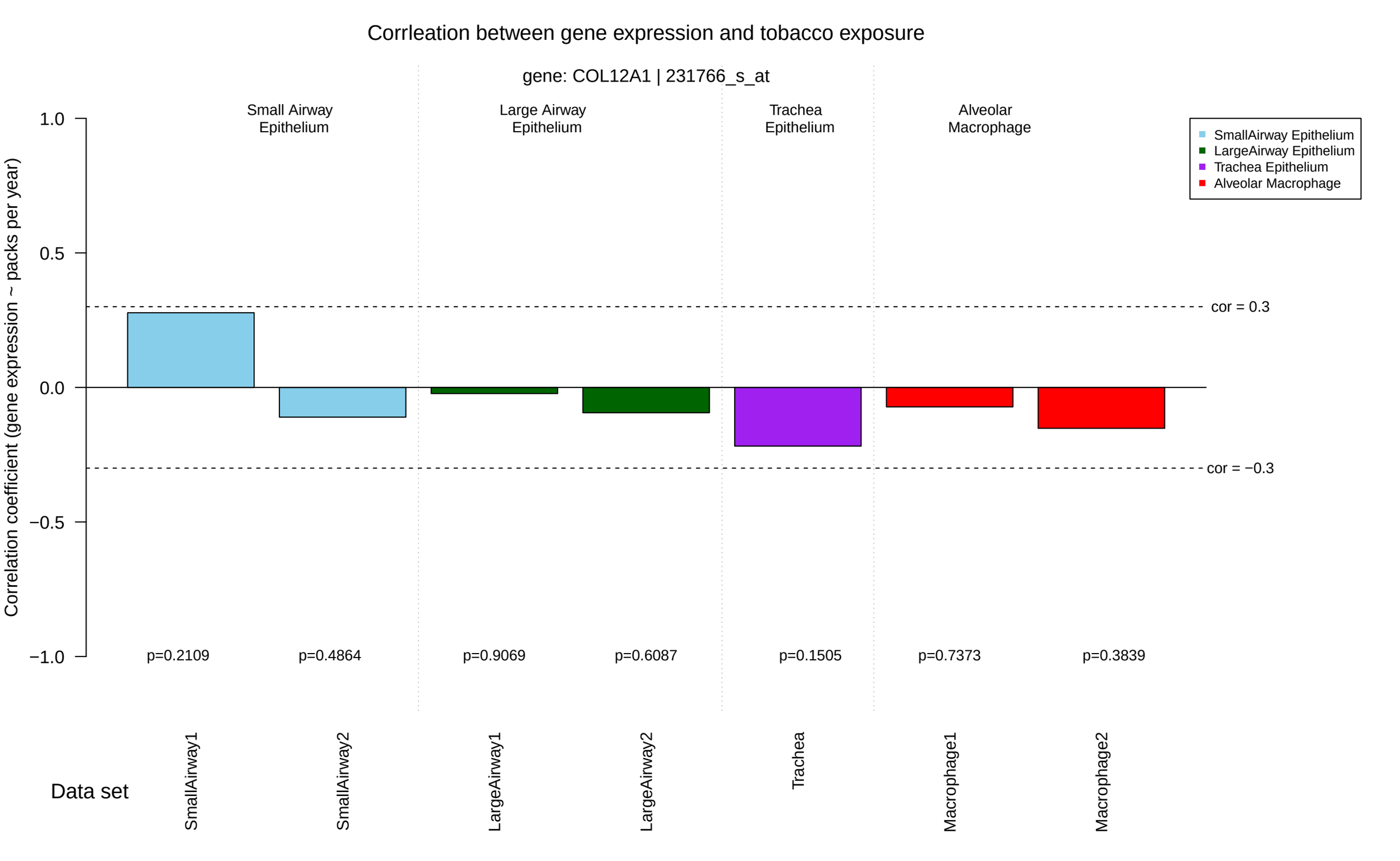

set: LargeAirway1 (p=0.4419)
COL12A1 | 231766_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6106)
COL12A1 | 231766_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0176)
COL12A1 | 231766_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8839)
COL12A1 | 231766_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1624)
COL12A1 | 231766_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL12A1 | 231766_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2109
p=0.4864
p=0.9069
p=0.6087
p=0.1505
p=0.7373
p=0.3839
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2